

Midterm Exam Sample

Question 1: Ridge Regression Solution

Given:

$$X = \begin{bmatrix} 1 & -1 \\ 1 & 2 \end{bmatrix}, \quad \lambda = 3, \quad y = \begin{bmatrix} 1 \\ 1 \end{bmatrix},$$

find the solution for ridge regression.

The ridge regression solution is given by:

$$\hat{\beta} = (X^T X + \lambda I)^{-1} X^T y.$$

Solution

Compute $X^T X$:

$$X^T X = \begin{bmatrix} 2 & 1 \\ 1 & 5 \end{bmatrix}.$$

Add λI :

$$X^T X + \lambda I = \begin{bmatrix} 5 & 1 \\ 1 & 8 \end{bmatrix}.$$

Invert:

$$(X^T X + \lambda I)^{-1} = \frac{1}{39} \begin{bmatrix} 8 & -1 \\ -1 & 5 \end{bmatrix}.$$

Compute $X^T y$:

$$X^T y = \begin{bmatrix} 2 \\ 1 \end{bmatrix}.$$

The ridge regression solution is:

$$\hat{\beta} = \begin{bmatrix} \frac{5}{13} \\ \frac{1}{13} \end{bmatrix}.$$

Question 2: PCA Reduction

Given $X = \begin{bmatrix} 2 & 0 \\ 0 & -1 \\ 1 & 1 \end{bmatrix}$, compute the centered data and the reduced version of the data using PCA. All computations must be shown using exact fractions.

Solution

Step 1: Compute the Mean of Each Column

The mean of each column is:

$$\text{Mean}_1 = \frac{2+0+1}{3} = 1, \quad \text{Mean}_2 = \frac{0-1+1}{3} = 0.$$

Thus, the centered data X_{centered} is:

$$X_{\text{centered}} = X - \text{Mean} = \begin{bmatrix} 2 & 0 \\ 0 & -1 \\ 1 & 1 \end{bmatrix} - \begin{bmatrix} 1 & 0 \\ 1 & 0 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ -1 & -1 \\ 0 & 1 \end{bmatrix}.$$

Step 2: Compute the Covariance Matrix

The covariance matrix is given by:

$$\Sigma = \frac{1}{2} X_{\text{centered}}^T X_{\text{centered}} = \frac{1}{2} \begin{bmatrix} 1 & -1 & 0 \\ 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ -1 & -1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & \frac{1}{2} \\ \frac{1}{2} & 1 \end{bmatrix}.$$

Step 3: Eigenvalue Decomposition

Next, we compute the eigenvalues and eigenvectors of the covariance matrix. The eigenvalues are:

$$\lambda_1 = \frac{1}{2}, \quad \lambda_2 = \frac{3}{2}.$$

The corresponding eigenvectors are:

$$v_1 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}, \quad v_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

Step 4: Projection onto the First Principal Component

The reduced version of the centered data projected onto the first principal component is:

$$X_{\text{reduced}} = X_{\text{centered}} v_1 = \begin{bmatrix} \frac{1}{2} \\ -\frac{1}{2} \\ 0 \end{bmatrix}.$$

Question 3: Kernel Ridge Regression

Given $k(x, z) = (x^T z)^2$, $X = \begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix}$, and $\lambda = 1$, solve the kernel ridge regression problem.

Solution

Step 1: Compute the Kernel Matrix

The kernel matrix K is:

$$K = \begin{bmatrix} 1 & 1 \\ 1 & 4 \end{bmatrix}.$$

Step 2: Compute $K^T K + \lambda K$

$$K^T K + \lambda K = \begin{bmatrix} 3 & 6 \\ 6 & 21 \end{bmatrix}.$$

Step 3: Compute $K^T y$

$$K^T y = \begin{bmatrix} 2 \\ 5 \end{bmatrix}.$$

Step 4: Compute α

The solution is:

$$\alpha = \begin{bmatrix} \frac{4}{9} \\ \frac{1}{9} \end{bmatrix}.$$

Question 4: Tucker Decomposition

Suppose we have a tensor X with frontal slices:

$$X_1 = \begin{bmatrix} 2 & 1 \\ -1 & 1 \end{bmatrix}, \quad X_2 = \begin{bmatrix} -1 & -1 \\ -1 & 1 \end{bmatrix}.$$

What is the U part of the SVD decomposition of $\mathcal{X}_{(3)}$?

Solution

Step 1: Mode-3 Unfolding of the Tensor

The mode-3 unfolding $\mathcal{X}_{(3)}$ is constructed as:

$$\mathcal{X}_{(3)} = \begin{bmatrix} 2 & 1 & -1 & 1 \\ -1 & -1 & -1 & 1 \end{bmatrix}.$$

Step 2: Compute $\mathcal{X}_{(3)}^T \mathcal{X}_{(3)}$

We compute:

$$\mathcal{X}_{(3)}^T \mathcal{X}_{(3)} = \begin{bmatrix} 5 & 4 \\ 4 & 4 \end{bmatrix}.$$

Step 3: Find the Eigenvectors of $\mathcal{X}_{(3)}^T \mathcal{X}_{(3)}$

The eigenvalues of $\begin{bmatrix} 5 & 4 \\ 4 & 4 \end{bmatrix}$ are $\lambda_1 = 9$ and $\lambda_2 = 0$.

The corresponding eigenvectors are:

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} -4 \\ 5 \end{bmatrix}.$$

Step 4: Normalize the Eigenvectors

After normalization, the U matrix is:

$$U = \begin{bmatrix} \frac{1}{\sqrt{2}} & -\frac{4}{\sqrt{41}} \\ \frac{1}{\sqrt{2}} & \frac{5}{\sqrt{41}} \end{bmatrix}.$$